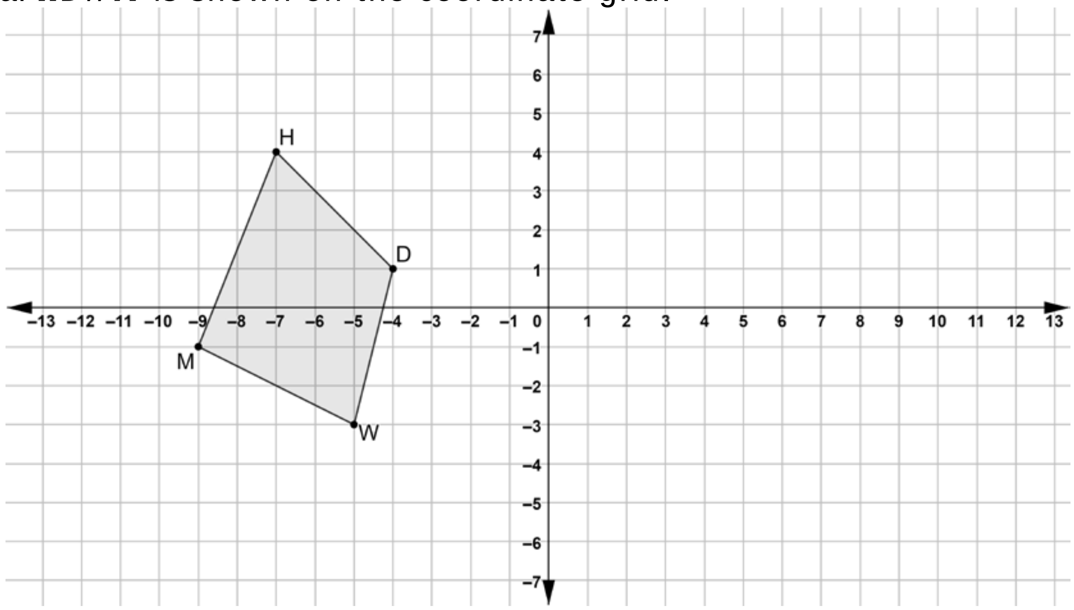
Geometry Name____________________
Unit 4 Date_____________________
Lesson 4 Homework Period___________________

Quadrilateral $HDWM$ is shown on the coordinate grid.



1. Draw the quadrilateral that is the result of a translation right 8 units and up 3 units and then reflected across the y-axis. Label the new quadrilateral $YBKT$.

Complete the table.

	Transformation	Mapping	Algebraic Description
2.		$HDWM \rightarrow H'D'W'M'$	
3.		$H'D'W'M' \rightarrow YBKT$	

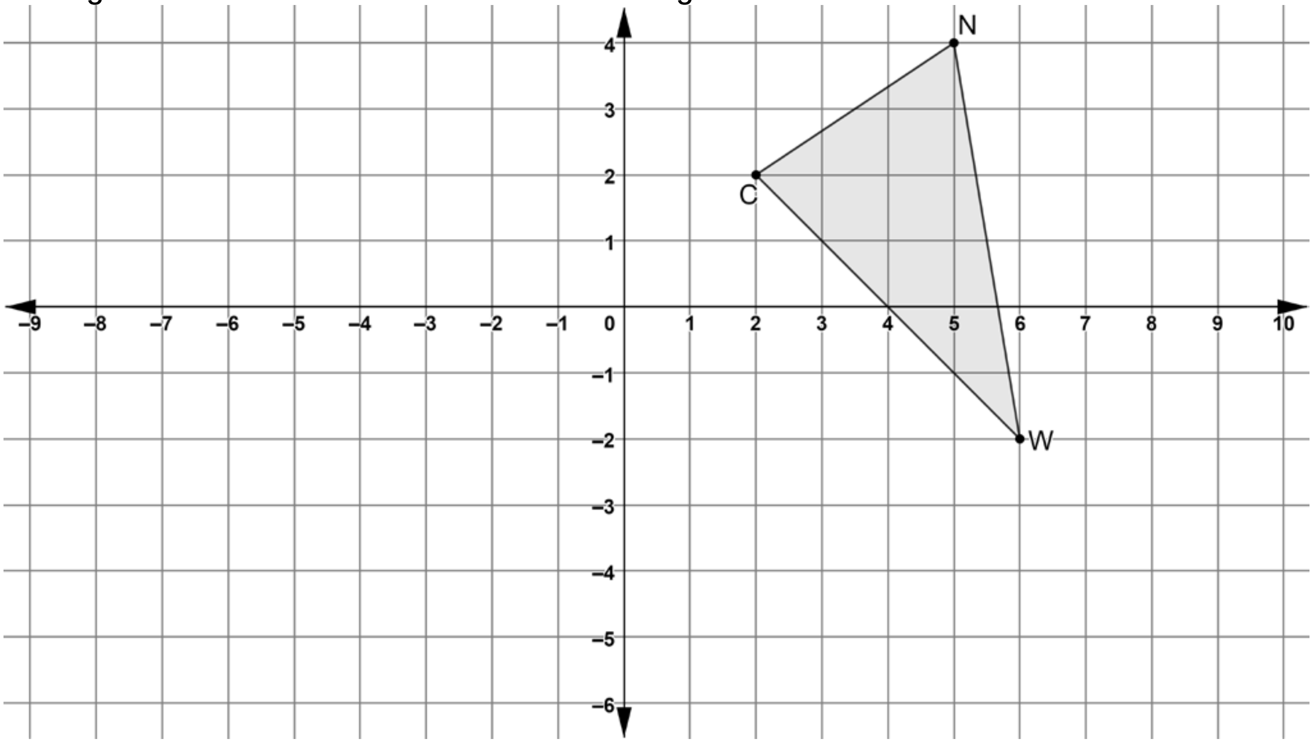
4. Complete the table of values for each quadrilateral.

Quadrilateral $HDWM$			Quadrilateral $H'D'W'M'$			Quadrilateral $YBKT$		
Vertex	x	y	Vertex	x	y	Vertex	x	y
H			H'			Y		
D			D'			B		
W			W'			K		
M			M'			T		

5. Explain how the tables for quadrilaterals $HDWM$ and $H'D'W'M'$ show the effect of the transformation on the coordinates that result in the algebraic description.

6. Explain how the tables for quadrilaterals $H'D'W'M'$ and $YBKT$ show the effect of the transformation on the coordinates that result in the algebraic description.

Triangle NWC is shown on the coordinate grid.



7. Draw the triangle that is the result of a rotation $90°$ clockwise about the origin and then a translation of left 3 units and up 4 units. Label the new triangle $N''W''C''$.

Complete the table.

	Transformation	Mapping	Algebraic Description
8.		$\triangle NWC \rightarrow \triangle N'W'C'$	
9.		$\triangle N'W'C' \rightarrow \triangle N''W''C''$	

10. Describe the connection between the algebraic description of the rotation and the vertices of $\triangle NWC$ and $\triangle N'W'C'$.